

Vision-Based Robotic Hot-Water Weeding for Precision Agriculture

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Abstract

Rising herbicide resistance, labour shortages, and sustainability concerns are driving interest in alternative weed control methods. This paper evaluates hot water as a robotic weed-control method and presents a modular hot-water weeder prototype developed for future open-field trials. A FarmBot-based feasibility study delivered water at 98°C to common weed species using different volumes, nozzle heights, and application frequencies. A single 150 ml treatment eliminated Brome (*Bromus sterilis*), Daisies (*Bellis perennis*), Dandelions (*Taraxacum officinale*), and Docks (*Rumex obtusifolius*), while Chickweed (*Stellaria media*) was eliminated after two treatments. Nettles (*Urtica dioica*) showed foliar injury but survived, and tillage radish (*Raphanus sativus*) showed no visible response. These results informed a modular multi-nozzle hot-water weeder using individually controlled nozzles, RGB vision, and YOLO-based weed detection to customise treatment volume by weed type. A 1.5 m FarmBot-mounted 16-nozzle prototype validated the targeting approach, and a mobile robot-mounted prototype extended the system to a 2 m, 32-nozzle boom with onboard water storage, heating, batteries, and ROS2 control. The vision system achieved 0.8736 precision, 0.742 recall, and 0.7611 mAP@0.5, while system testing achieved an 86.67% integrated detection-and-treatment success rate with a 6.67% false-positive rate. The results demonstrate that hot water can be selectively applied using low-cost RGB vision and modular actuation as a maintainable non-chemical approach to robotic precision weeding.

Keywords: Precision weeding, hot water weeding, agricultural robotics, machine vision, YOLO

1 Introduction

Weeds are one of the largest yield constraints in crop production and can act as vectors for disease and parasites. Reliance on herbicides is becoming unsustainable due to increasing resistance and regulatory constraints linked to ecological risk [Duke, 2017, Bamal et al., 2024, Murdoch, 2018]. Manual weeding is ecologically favourable but limited by labour cost and availability [Win et al., 2025], while mechanical weeding can disturb soil, increase erosion and greenhouse gas emissions, disrupt soil microbiota, and damage nearby crops. Precision weeding addresses these limitations by using machine vision and automation to target individual weeds, reducing chemical use and crop or soil disturbance [Slaughter et al., 2008]. As herbicide development becomes more expensive and robotic systems become more affordable, precision approaches are becoming viable tools for sustainable crop production [Upadhyay et al., 2024, Xie et al., 2025].

Integrated Weed Management (IWM) combines prevention, cultural methods, limited targeted chemical use, and robotic weeding. Preventing weed-seed dispersal through machinery hygiene, using rotation, cover



Figure 1: Hot water weeding evaluation.

crops and polyculture, and targeting individual weeds can reduce herbicide dependence and environmental contamination [Murdoch, 2018, Duke, 2017]. Hot water is attractive within this framework because it requires no external chemical agent, does not disturb the soil, has lower fire risk than flame weeding, and can also be adapted for targeted irrigation or fertigation. The main contributions are: (i) hot-water treatment results for common weed species; (ii) a YOLO-based vision-to-actuation pipeline for nozzle-level treatment; (iii) a 16-nozzle FarmBot validation prototype; and (iv) a 32-nozzle mobile robot-mounted prototype developed for future open-field trials.

2 State of the Art

2.1 Alternative Sustainable Weeding Methods

Mechanical weeding is one of the oldest non-chemical approaches and remains common in organic cereal systems [Quan et al., 2022]. Spring tine harrows and rotary hoes are widely used [Lötjönen and Mikkola, 2000], but they do not target weeds individually and may damage crop plants, particularly near stems or within rows [Lötjönen and Mikkola, 2000]. Reviews of weeding strategies highlight the need for practical, maintainable technologies [Bloomer et al., 2024], and farmer surveys show a preference for modular tools that can be repaired on farm with minimal external support [Hameed et al., 2025].

Other non-chemical treatments include vinegar, laser, and thermal weeding. Wood vinegar can inhibit seed germination at low concentrations, but repeated application, variable species response, and uncertain ecological effects remain limitations [Zhang et al., 2022]. Laser systems such as WeLASER provide precise non-contact treatment but face barriers including high cost, energy demand, fire risk, and maintenance complexity [Andreasen et al., 2024]. Thermal weeding damages plants through protein denaturation and membrane rupture [Peerzada and Chauhan, 2018]. Hot water has shown particular promise because it avoids soil disturbance, reduces fire risk, and causes less surface wear than mechanical or flame-based methods [Hansson, 2002, Sarıtaş and Gürbüz, 2024]. Astatkie and Havard [T. Astatkie, 2007] found hot water effective against young weeds at low speeds and consistent across weather conditions without drift or run-off. Martelloni et al. [Martelloni et al., 2021] showed that hot water reduced regrowth and biomass within 30 days, while noting that hot foam can achieve similar effects with lower doses. However, hot water remains simpler to deploy, requires no foaming agent, and is well suited to precision application.

2.2 Weed Detection and Targeting

Precision weeding requires reliable identification of individual weeds. 2D RGB vision systems are widely used because they are low cost, accessible, and compatible with many detection models. They distinguish weeds and crops using colour, size, morphology, and position [Guzel et al., 2024]. Traditional machine vision works best when plants are visually separated [Slaughter et al., 2008], but 2D imaging can struggle under occlusion [Xie et al., 2023]. Potena et al. [Potena et al., 2016] showed that infrared thermography can improve segmentation under difficult lighting, but at increased sensing cost and complexity. Quan et al. [Quan et al., 2022] demonstrated RGB-based crop–weed detection for intra-row maize weeding using a mechanically actuated tool, illustrating both the potential of vision-guided weeding and the complexity of mechanical end effectors.

Vision-guided spraying offers a simpler actuation route. Terra et al. [Terra et al., 2021] linked real-time weed detection to electronically controlled nozzles for site-specific application, while Wu et al. [Wu et al., 2019] showed that multiple RGB cameras can broaden detection coverage. Similar vision-based selective spraying systems have demonstrated the potential to reduce herbicide use by treating only detected weed locations [Raja et al., 2023]. Commercial systems such as Verdant Robotics' SharpShooter [VerdantRobotics, 2025] demonstrate the potential of targeted application and report large reductions in chemical and labour use, but their sophistication may raise concerns around maintainability and durability. This motivates simpler modular systems for field robotics.

3 My Work

3.1 Validating Hot Water for Weeding

3.1.1 Experiment Setup

To validate boiling water as a cereal-crop weed control method, an experiment studied the effect of water heated to 98°C on common weeds while varying volume, application frequency, and nozzle height. Consultation with local farmers identified Brome, Chickweed, Nettles, grass weeds, and tillage radish as common in European temperate-zone cereal crops. A 3×2 m mown grass area, containing approximately 75% fescue and 25% perennial rye, was also included to investigate Docks, Daisies, and Dandelions for grassland or golf-course-style applications. Rows of oats were planted between weeds to observe crop effects.

The platform was a FarmBot Genesis XL installed on a raised greenhouse bed. It was modified with an in-line water boiler, heat-resistant piping installed in parallel with the cold-water lines, and a solenoid valve for hot-water release. Weeds were planted by the robot in a known coordinate system and received frequent light watering. The FarmBot was controlled using an open-source ROS2 interface. In total, 26 plants of each species were planted, allowing each treatment parameter to be applied to two plants. The same treatment schedule was applied within each tested species group, as shown in Table 1. Some parameters followed [Hansson and Mattsson, 2002]. Plants were imaged using a Luxonis OAK-D Lite before treatment and seven days after treatment, providing both health records and a dataset for future weed-detection models.

Plant No.	Frequency	Nozzle Height	Volume
P1	Control	-	-
P2	Every Three Weeks	18 cm	300 ml
P3	Every Three Weeks	18 cm	150 ml
P4	Every Three Weeks	9 cm	300 ml
P5	Every Three Weeks	9 cm	150 ml
P6	Every Two Weeks	18 cm	300 ml
P7	Every Two Weeks	18 cm	150 ml
P8	Every Two Weeks	9 cm	300 ml
P9	Every Two Weeks	9 cm	150 ml
P10	Every Week	18 cm	300 ml
P11	Every Week	18 cm	150 ml
P12	Every Week	9 cm	150 ml
P13	Every Week	9 cm	300 ml

Table 1: Spray treatment parameters applied to each plant within each tested species group.

3.1.2 Experiment Results & Discussion

The six-week experiment showed clear species-dependent responses. Brome was completely eradicated under all treatment conditions. Chickweed required two applications and would likely benefit from a broader spray pattern to improve full-plant coverage. Tillage radish (*Raphanus sativus*) showed no visible response, indicating high tolerance. Nettles (*Urtica dioica*) sustained foliar injury but survived due to robust root systems, suggesting that earlier and higher-volume treatments may improve control. Dandelions, Daisies, and Docks were effectively killed, including damage to large taproots, though higher water volumes caused collateral grass damage. Neighbouring oat crops were unaffected and produced a strong yield. Overall, with the exception of Nettles and tillage radish, the tested species were controlled using relatively low volumes and low application frequency. Nozzle height had little effect on efficacy, and the results suggest that hot water is most effective on younger weeds before extensive root systems develop.

3.2 Precision Weeder Design & Implementation

3.2.1 Computer Vision

The computer vision system was developed to detect weeds, classify them by species, and provide spatial information for nozzle-level hot-water treatment. Unlike a standalone image-classification task, the output of the vision model had to be suitable for robotic actuation. Each detection therefore needed to provide a weed

class, confidence score, and image-space location that could be converted into a centroid and passed to the control pipeline.

The dataset formed during the hot-water validation experiment was expanded using a third-party dataset [AUENG Vision Group, 2025]. This increased variation in lighting conditions, soil types, plant growth stages, and image backgrounds. Images collected during the FarmBot experiment were particularly useful because they represented the same camera perspective, crop context, and treatment environment used during system validation. The combined dataset was labelled using bounding boxes, as this format is directly compatible with YOLO-based object detection and provides a simple method for extracting weed centroids for downstream robotic control.

A YOLOv8n model was trained using mAP, precision, and recall as evaluation metrics. Image augmentations including Flip, Crop, Mosaic, and Shear were applied to improve robustness to variation in weed orientation, scale, partial visibility, and field background. The best-performing model checkpoint was selected for inference. The trained model achieved 0.8736 precision, 0.742 recall, and 0.7611 mAP@0.5. This performance showed a useful balance between minimizing false positives, which could result in crop damage, and maintaining sufficient true positive detections for practical weed treatment. Training curves showed steady reductions in classification and bounding-box losses, while validation losses fluctuated slightly due to dataset variability but remained stable overall.

Compared with existing RGB crop–weed detection work, the achieved YOLOv8n precision of 0.8736 and mAP@0.5 of 0.7611 show that the detector provides a practical baseline for real-time robotic treatment. Direct numerical comparison is limited because weed-detection results are strongly affected by species, crop type, camera viewpoint, lighting conditions, dataset composition, and evaluation protocol. However, unlike mechanically actuated intra-row systems [Quan et al., 2022], the proposed approach links RGB detections to selective nozzle actuation, and unlike conventional site-specific spraying systems [Terra et al., 2021, Raja et al., 2023], the treatment medium is hot water rather than herbicide.

For robotic use, the detector output was converted into a treatment decision rather than being used only as an image-level result. Each bounding box was converted to a centroid, associated with its predicted weed class and confidence score, and passed to the ROS2 control pipeline. The centroid position was then related to the camera frame and used to estimate whether the weed lay within the effective spray corridor. Detections outside the workable region were ignored, while valid detections were tracked as the robot moved forward. This allowed the system to convert image-space detections into nozzle-specific treatment commands.

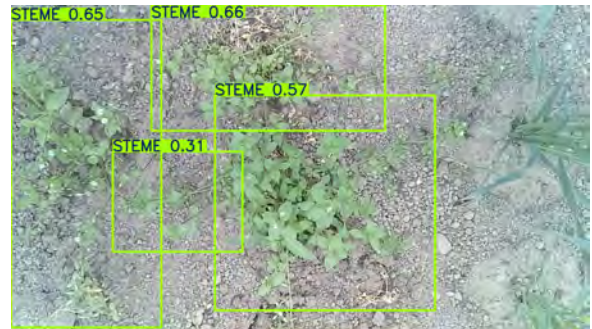


Figure 2: Custom trained weed detection model validated on Chickweed and oat crop.

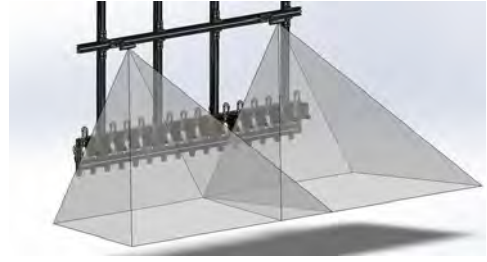
The use of bounding-box detection was chosen as a practical compromise between localization accuracy and computational efficiency. Pixel-level segmentation could improve localization where plants overlap, but it would also increase annotation effort and processing requirements. For the current nozzle spacing and spray pattern, bounding-box centroids provided sufficient spatial information for initial prototype validation. Future work will expand dataset diversity across species, growth stages, lighting conditions, and outdoor field environments. Segmentation may also be introduced later to improve targeting where weeds and crops overlap or where the weed canopy extends beyond the bounding-box centre.

3.2.2 Robot Design

A 1.5 m FarmBot-mounted sprayer array was first developed to validate the vision-guided targeting method. The array used 16 individually controlled spray nozzles mounted to an aluminium frame. Each nozzle was regulated by a solenoid valve actuated through two 8-channel relay modules and a 16-channel PWM driver, in-



(a) FarmBot Prototype.



(b) FarmBot CAD.



(c) Mobile robot CAD.



(d) Robot-mounted prototype.

Figure 3: Development from the FarmBot validation prototype to the robot-mounted prototype.

terfaced with a Raspberry Pi 4B through a custom daughterboard. Two Luxonis OAK-D Lite cameras provided image input, while a networked computer ran the YOLOv8n model and returned detected weed centroids. The validation assembly is shown in Fig. 3.

A mobile robot-mounted prototype was then implemented as a hot-water weeding tool. The boom was scaled to 2 m to match the mobile robot's track width and fitted with 32 nozzles, improving targeting resolution. It was mounted to the three-point hitch using custom Category 1 steel mounts, with the lower arms carrying the vertical load and the top link setting boom pitch.

The onboard prototype system included a 200 L water tank, two 24 V LiFePO₄ batteries, a 2000 W inverter, a 1500 W instant boiler, a 24 V diaphragm pump, an immersion heater, and high-temperature silicone tubing feeding the 32-nozzle manifold. Water was drawn from the tank, preheated, passed through the boiler, and delivered to the boom above 95°C under typical operating flow. For the mobile robot-mounted prototype, an OAK-D PoE camera and the robot's onboard NVIDIA Orin replaced the external FarmBot computer, while ROS2 nodes handled camera input, inference, weed tracking, odometry, and nozzle control.

3.2.3 Control Architecture

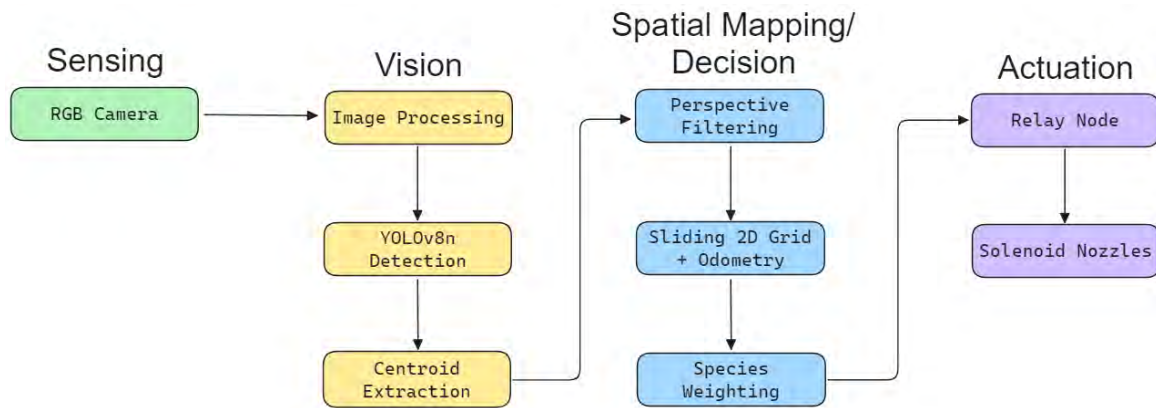


Figure 4: System architecture linking RGB weed detection to nozzle-level hot-water actuation.

The ROS2 control architecture, shown in Fig. 4, is modular and platform independent. Communication with the main robot is limited to master deactivation, initialization configuration, and odometry. Each camera

has its own ROS run node and processing node, with both processing nodes communicating independently with the relay control node. The processing node contains image processing, model inference, and interpretation. The odometry node provides distance travelled, allowing weed positions to be represented relative to the nozzle array over time.

Due to camera perspective, some visible weeds are outside the sprayer trajectory shown in Fig. 5. These are removed using trigonometric height-wise planar weed localization based on the camera frustum, enabling real-time localization without repeated camera recalibration. The height-wise variable computation and nozzle mapping are expressed as:

$$l_p = 2h \tan \frac{FOV_h}{2}, \quad L_p = 2\sqrt{h^2 + (h \tan(FOV_v))^2} \tan \frac{FOV_h}{2}, \quad l' = \frac{l}{L} Y_{cam} \quad (1)$$

$$\forall P_i(x_i, y_i): \quad (a+b) = \frac{l'}{2} + \frac{y_i}{y_{cam}} \left(\frac{X_{cam}}{2} - \frac{l'}{2} \right), \quad b = \left| x_i - \frac{X_{cam}}{2} \right|$$

If $b > a+b$, ignore detection. Otherwise:

$$X_r = \begin{cases} \frac{X_{cam}}{2} - \frac{b}{a+b} \frac{X_{cam}}{2}, & x_i < \frac{X_{cam}}{2} \\ \frac{X_{cam}}{2} + \frac{b}{a+b} \frac{X_{cam}}{2}, & \text{otherwise} \end{cases} \quad (2)$$

Here, h is camera height, FOV_h and FOV_v are camera fields of view, and X_{cam} and Y_{cam} are image dimensions. $P_i(x_i, y_i)$ is the detected weed centroid, X_r is the nozzle-mapping coordinate, l_p , L_p , l' , l , and L are perspective-scaling terms, and a and b define the spray-trajectory acceptance region.

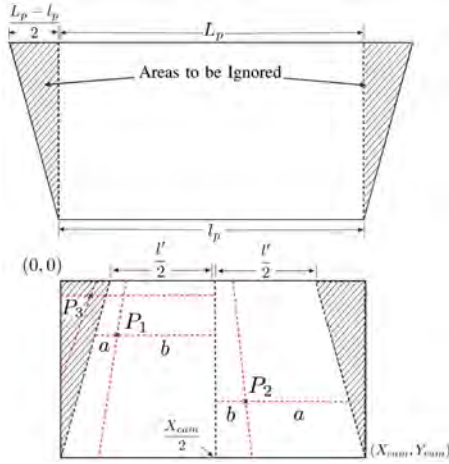


Figure 5: Camera ground footprint and corresponding image frame. P_3 is ignored as it lies outside of the nozzle trajectory

A weed prediction score was computed by combining detection confidence with observation duration. As the robot moved, weeds were tracked toward the nozzle array using a sliding 2D grid with 8 columns per camera and a variable number of perspective-scaled rows. When a weed was detected, its score was assigned to the relevant cell. As odometry advanced, lower rows were popped and new upper rows added. If a detection exceeded a calibrated threshold, the nozzle corresponding to that row was activated once popped. The score was further weighted by weed species so valves could remain open for different durations and deliver species-specific volumes aligned with the hot-water experiment. Because no pressure control was present, simultaneous nozzle activation reduced pressure per nozzle; this was considered during treatment timing.

3.3 Precision Weeder Evaluation

Four experiments evaluated precision, accuracy, efficacy, and reliability. Targeting precision was calculated by comparing the centre point of a weed with the centroid of the surrounding wet area produced by the weeder. Measurements from Brome, Chickweed, and Nettle are shown in Table 2, where M1–M3 are repeated offset measurements.

Weed	M1	M2	M3	Mean	Std. Dev.
Brome	27	35	8	23.3	11.2
Chickweed	15	50	25	30.0	15.9
Nettle	25	60	40	41.7	14.6

Table 2: Targeting offset measurements in mm.

Across species, the mean offset was 31.7 mm with an average standard deviation of 13.9 mm. Compared to the FarmBot nozzle spacing of 93.75 mm, these offsets show that the system targeted the correct region, but also that a single nozzle may not cover all centroid positions. Finer nozzle resolution is therefore beneficial and motivated the 32-nozzle robot-mounted boom.

In a controlled test, the system detected and treated 26 out of 30 weeds, giving an 86.67% integrated detection-and-treatment success rate. This provides a system-level result, while the YOLOv8n precision value of 0.8736 reports model-level detection performance. Reliability was also demonstrated, as only one of 15 crop plants, including tomato, oats, barley, and wheat, was mistakenly targeted, corresponding to 93.33% crop protection success and a 6.67% false-positive rate. These results indicate strong baseline accuracy and reliability, with potential improvement through expanded training data and inclusion of non-weed detections.

Dose delivery was tested by directing a nozzle into a measuring jug as the weeder advanced toward a weed. The system was calibrated to deliver at least 150 ml to Brome and Chickweed and 300 ml to Nettle. The 150 ml dose ranged from 130–190 ml with a standard deviation of 28.3 ml, while the 300 ml dose ranged from 250–390 ml with a standard deviation of 60.6 ml. Variation was mainly attributed to nozzle tolerances, adjustment, and pressure reduction during simultaneous actuation. Overall, the robot-mounted prototype extends the FarmBot validation system to a mobile platform with onboard water, heat, power, sensing, and ROS2 control.

4 Conclusion, Limitations and Future Work

This paper investigated hot water as a weed-control method for future open-field robotic application. A gantry-based robot applied water at 98°C to weeds common in temperate cereal farms and grassland environments. Brome, Daisies, Dandelions, and Docks were controlled after one 150 ml treatment, Chickweed required two treatments, Nettles survived due to robust root systems, and tillage radish showed no visible response. A 1.5 m FarmBot-mounted precision weeding tool was then evaluated using 16 electronically controlled nozzles and a vision system trained for Brome, Chickweed, and Nettle detection. The system achieved an 86.67% integrated detection-and-treatment success rate, 93.33% crop protection success, and species-specific dose delivery. The design was extended to a mobile robot-mounted prototype with a 2 m, 32-nozzle boom, onboard water storage, heating, power, sensing, NVIDIA Orin inference, and ROS2 control.

The main limitations are targeting resolution, pressure variation during simultaneous nozzle activation, operating speed, energy demand, and model robustness under diverse field conditions. Future work will include open-field trials on the mobile robot, evaluation of the 32-nozzle system, improved heating and pressure control, expanded training data, and workshops with farmers to assess adoption, maintainability, and usability.

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